

# Navigating Through Fear and Greed

The Experience-Driven Disposition Effect

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SFS Cavalcade 2026

# The investor's worst enemy

*"The investor's chief problem – and even his worst enemy – is likely to be himself."*

– Benjamin Graham

## The disposition effect.

Investors hold losers too long and sell winners too soon – documented across countries, asset classes, periods, and institutional investors. Shefrin & Statman 1985; Odean 1998

## Why care?

- The stocks investors *sell* outperform the ones they *hold* by about 3.4 percentage points per year Odean 1998 – the “it’ll bounce back” rationale isn’t in the data.
- The decision hinges on the investor’s own purchase price – not on where the stock is going. Tax planning prescribes the opposite: sell losers, hold winners Constantinides 1984.

## Open question.

Does *experience* attenuate it – and if so, do gains and losses teach the same lesson?

# The experience puzzle

## One view: experience teaches rationality.

Field markets and lab studies show behavioral biases **shrink** with trading activity.

List (2003, 2004); Seru, Shumway & Stoffman (2010); Dhar & Zhu (2006).

## Another view: experience amplifies bias.

Bad experiences – market crashes, big losses – can **entrench** or even **worsen** biases.

Malmendier & Nagel (2011); Choi et al. (2007); Kuhnen (2015).

## Goal of this paper:

separate *winning* from *losing* experiences and ask whether they shape the disposition effect symmetrically.

## The tension.

Existing studies measure experience as **cumulative trade count** – agnostic about *what was learned*.

But realized gains and realized losses are emotionally and informationally different.

## Sample selection.

Many prior datasets cover **bull-only** periods – so they over-sample winners and under-sample losers.

A boom → bust window is needed to identify the asymmetry.

- **Data.** Account-level transactions for **189,530 Chinese retail investors**, **40M+ orders**, **Jul 2013 – Feb 2016**; covers the full boom (SSE: 2,000 → 5,000) and bust (5,000 → 2,700).
- **Empirical result – the asymmetry.** Past trading experiences reshape the disposition effect, but **valence matters**:
  - **greed**: sequences of salient *gains* **attenuate** the disposition effect.
  - **fear**: sequences of salient *losses* **amplify** it.
- **Mechanism.** A disciplined **memory-based cued-recall model** (Howard & Kahana 2002; Kahana et al. 2025) – combines temporal context with stock-return skewness to generate the baseline disposition effect *and* its asymmetric experience response from a single mechanism.
- **Why it matters.** Static utility-based accounts (prospect theory, realization utility) predict a **fixed** DE per investor – and lack a principled reason for the **sign-flip** (gains ↓ DE, losses ↑ DE). Memory dynamics deliver it from one channel.

1. **Data and measurement: Chinese boom–bust, experience counts**
2. The asymmetric experience-driven disposition effect
3. Robustness: thresholds, holding period, mechanical bias
4. Mechanism: cued-recall memory model
5. Heterogeneity, wide framing, and takeaways

# Setting: a full boom–bust cycle

## Why this window?

Most prior disposition-effect papers use *bull-only* samples. We span bull, crash, and recovery – so investors accumulate *both* salient wins and salient losses.



SSE Composite Index. Green shading: our sample window (Jul 2013 – Feb 2016). Grey: 2015 crash.

## Source.

A major Chinese brokerage – complete records from account opening: every buy, every sell, prices, volumes, account demographics, and a survey-based risk-tolerance score.

## Coverage.

- 189,530 retail investors
- ~4,000 listed A-share stocks
- 40,000,000+ transactions
- 43,407,019 investor-stock-day observations
- Average tenure: 340 days

## Investor characteristics (means).

|                            |            |
|----------------------------|------------|
| Male                       | 55%        |
| Age                        | 38 years   |
| Risk-taking (1–5)          | 3.05       |
| Education (1–7)            | 4.11       |
| <hr/>                      |            |
| Stocks per portfolio       | 7.6        |
| Portfolio value (median)   | RMB 63,472 |
| Losing experiences (>5%)   | 6.45       |
| Winning experiences (>5%)  | 5.66       |
| Losing experiences (>30%)  | 0.68       |
| Winning experiences (>30%) | 0.28       |

## Bottom line.

A large, granular, demographically diverse field sample spanning a full cycle.

# Measuring experience and the disposition effect

## Key innovation.

We *separate* past wins from past losses – and let each have its own slope.

### Position & trading episode.

A *position* starts at the first purchase of a stock; ends at the first sale (partial sales start a new episode).

### Significant experiences.

For threshold  $X \in \{5\%, 10\%, \dots, 30\%\}$ :

$$\text{LossExp}_{i,t} = \sum_{T < t} \mathbf{1}\{r_{i,T} < -X\}$$

$$\text{WinExp}_{i,t} = \sum_{T < t} \mathbf{1}\{r_{i,T} > +X\}$$

Salience cutoff captures outcomes large enough to be *encoded in memory* (and net of trading costs).

### Disposition effect.

$$\text{PGR} = \frac{\text{realized gains}}{\text{realized gains} + \text{paper gains}}$$

$$\text{PLR} = \frac{\text{realized losses}}{\text{realized losses} + \text{paper losses}}$$

$$\text{DE} = \text{PGR} - \text{PLR} > 0 \Rightarrow \text{bias.}$$

### In our sample.

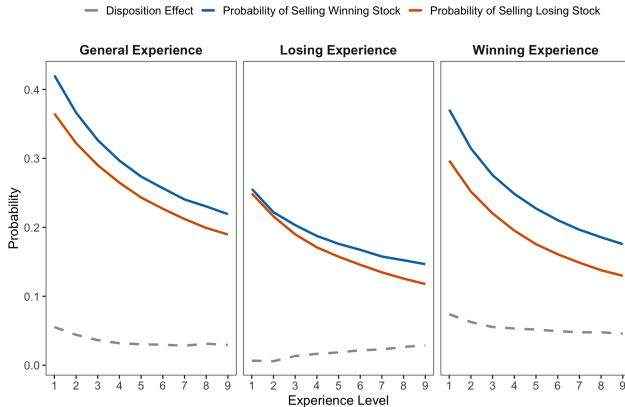
12.4% of gains realized; 8.3% of losses realized;  
 $\Rightarrow \text{DE} \approx 4.1$  p.p. Right in line with Odean (1998),  
Seru et al. (2010).



# A first look: experience reshapes selling, asymmetrically

## The dashed line tells the story:

DE *shrinks* along the winning-experience axis; *rises* along the losing-experience axis.



$x$ -axis: count of prior trading episodes of each type. Solid:  $P(\text{sell}|\text{winner})$  and  $P(\text{sell}|\text{loser})$ . Dashed: DE = gap.

## Baseline: trading experience attenuates DE

$$\text{Sale}_{i,j,t} = \alpha + \beta_1 \text{Gain}_{i,j,t} + \beta_2 \text{Exp}_{i,t} + \beta_3 (\text{Gain}_{i,j,t} \times \text{Exp}_{i,t}) + \alpha_i + \lambda_j + \tau_t + \varepsilon.$$

| Dep. var.: Sale $\times 100$ | (1) No FE | (2) +Inv&Stock FE | (3) +Time FE |
|------------------------------|-----------|-------------------|--------------|
| Gain                         | 4.49***   | 4.10***           | 2.75***      |
| Experience (all trades)      | -0.010*** | -0.018***         | 0.003**      |
| Gain $\times$ Exp            | -0.0043** | -0.0046***        | -0.0020**    |
| Observations                 |           | 43,388,866        |              |

**Read.** The Gain coefficient ( $\approx 2.75$ ) is the disposition effect. The negative interaction means: *the more you've traded, the smaller your DE* – consistent with Seru, Shumway & Stoffman (2010).

**So far, just learning. But this measure doesn't say *what* was learned.**

# Main result: fear amplifies, greed attenuates

Split experience by valence at the  $|X| > 5\%$  threshold:

$$\text{Sale}_{i,j,t} = \alpha + \beta_1 \text{Gain} + \beta_2 \text{LossExp} + \beta_3 \text{WinExp} + \beta_4 \text{Gain} \times \text{LossExp} + \beta_5 \text{Gain} \times \text{WinExp} + \text{FE} + \varepsilon.$$

| Dep. var.: Sale $\times 100$      | (1) Loss only | (2) Win only | (3) Joint, $ X  > 5\%$ | (4) Joint, $ X  > 30\%$ |
|-----------------------------------|---------------|--------------|------------------------|-------------------------|
| Gain                              | 2.67***       | 2.66***      | 2.67***                | 2.43***                 |
| Loss experience                   | 0.016***      |              | 0.003                  | 0.066                   |
| Win experience                    |               | 0.010        | 0.003                  | 0.015                   |
| Gain $\times$ LossExp <b>fear</b> | 0.0007        |              | +0.0100***             | +0.0907***              |
| Gain $\times$ WinExp <b>greed</b> |               | -0.0068***   | -0.0103***             | -0.0634*                |
| Observations                      |               |              | 43,388,866             |                         |
| $R^2$                             | 0.138         | 0.138        | 0.139                  | 0.139                   |

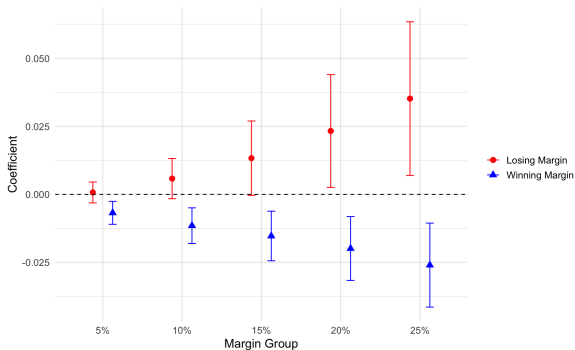
**Magnitudes.** At the 5% cutoff, a 2-sd rise in salient *losses* raises the DE by  $\approx 10\%$ ; a 2-sd rise in salient *wins* attenuates it by  $\approx 10\%$ . At 30%, the asymmetry is roughly 9 $\times$  larger per experience.

**The asymmetry only shows up once both terms are entered jointly – because winners and losers come bundled in real portfolios.**

# Asymmetry strengthens with salience

**The wider the cutoff, the larger the gap.**

If the asymmetry were a noise artifact, raising the threshold would weaken it. Instead, it sharpens.



$\hat{\beta}_4$  (red, Gain  $\times$  LossExp) and  $\hat{\beta}_5$  (blue, Gain  $\times$  WinExp),  $\pm$  90% CI, across  $X \in \{5\%, 10\%, 15\%, 20\%, 25\%\}$ .

Only *salient* outcomes get encoded; small wins and losses are noise.

# It's not portfolio composition or a V-shape

## Worry 1. Mechanical portfolio arithmetic.

Selling a loser tilts the rest of the portfolio toward winners  $\Rightarrow$  does the next trade just *look* like more DE?

We run 200 Monte Carlo experiments under three null DGPs (random; fixed DE; fixed DE + V-shape). **None** reproduces our  $(+0.010, -0.010)$  pair – not even directionally.

► See MC

## Worry 2. Experience share vs. count.

Reframe experience as the *share* of trades that were big losses

(wins).  $\text{Gain} \times \text{LossShare} = +32.5^{***}$ ;

$\text{Gain} \times \text{WinShare} = -6.8^{***}$ . Asymmetry sharpens.

## Worry 3. Holding period / V-shape.

Ben-David & Hirshleifer (2012) show a contemporaneous V-shape in  $P(\text{sell} \mid \text{paper return})$ . Could our experience gradient just relabel that?

Re-estimate within holding-period bins  $(0, 60]$  and  $(61, 250]$  days. Selling slope falls with age – but the *experience* interactions retain sign, significance, and magnitude.

## Worry 4. Unobserved skill / portfolio gains.

Add Portfolio Gain interactions à la An (JF 2024). Asymmetric experience effect survives at the portfolio level.

**The asymmetry is not a portfolio-arithmetic, sample-selection, holding-period, or skill artifact.**

# What can – and can't – explain the asymmetry

## Static preference accounts.

- Prospect theory + reference points.
- Realization utility (Barberis & Xiong).
- Loss-averse CARA / CRRA.

All deliver a baseline DE with fixed parameters – but *learning shifts parameters*, and *why* parameters would shift **asymmetrically** from gains vs. losses is not pinned down.

## Belief-updating accounts.

The lab literature is itself *internally divided*: **Kuhnen (2015)** finds loss-domain **pessimism** (which, applied to selling, predicts DE *attenuation* on the loss side – opposite of what we find); **Eil & Rao (2011)** find the *opposite* asymmetry for self-relevant signals; **Coutts (2019)** finds no asymmetry at all without self-relevance. No single belief-updating story delivers our **sign-flip**.

## This paper's proposal.

The decision to sell is a **cued recall** task.

The cue is “stocks I own.” What pops to mind depends on:

- **frequency** of past {stock, return} co-occurrences;
- **contextual similarity** of past trades to the present decision;
- the underlying **skewness** of returns.

## Building blocks.

Temporal-context model of Howard & Kahana (2002); recall-vs-recognition formalisation in Kahana et al. (2025).

# A cued-recall model of selling

**Primitives.** Each episode is a feature–context pair  $(f_i, x_i)$  stored as outer products in a memory matrix

$$M_n = M_{n-1} + x_i f_i^\top.$$

Features are basis vectors – e.g., Owned, Sold, Large, Small, Other.

**Cued recall.** Cue  $f_{\text{cue}} = \text{“stocks I own.”}$  Recover context  $x_{\text{cue}} \propto M f_{\text{cue}}$ , then recover features  $\tilde{f} \propto M^\top x_{\text{cue}}$ . A feature pops if it has nonzero weight *and* its context matches the cue.

**Recall probability** (Kahana et al. 2025):

$$\mathbb{P}(\text{recall Large} \mid \text{Owned}) \propto \frac{n_{\text{Owned, Large}}}{\sqrt{n_{\text{Owned}} n_{\text{Large}}}}.$$

**Behavioral rule.** The investor *sells the stock whose return pops to mind*.

► Math details

## Why this delivers the disposition effect

Large returns occur almost exclusively for *stocks*; small movements are everywhere (“other” commodities, prices in shops). Because stock returns are **positively skewed** (paper-return skewness = 23.1 in our sample), *large gains* dominate the recall set, so the cue “stocks I own” pulls up gains  $\Rightarrow$  investors sell winners. Where Large = “large return,” Small = “small return.”

# The model reproduces the asymmetric experience effect

## Three theorems (informal).

1. **Baseline DE.** With sufficiently many “other” small returns and positive return skewness,  $\mathbb{P}(\text{recall Large}) > \mathbb{P}(\text{recall Small})$ , so **the investor sells winners more than losers.**
2. **Winning experiences.** Selling  $k$  winners re-tags those gains as Sold rather than Owned:

$$n_{\text{Owned}, \text{Large}} \mapsto n_{\text{Owned}, \text{Large}} - k, \quad n_{\text{Sold}, \text{Large}} \mapsto n_{\text{Sold}, \text{Large}} + k.$$

$\Rightarrow \mathbb{P}(\text{Large} \mid \text{Owned})$  falls  $\Rightarrow$  **greed**: the DE **attenuates**.

3. **Losing experiences.** Each realised loss shifts that loss out of Owned into Sold. What’s left in “Owned” is even more dominated by large gains:

$$\mathbb{P}(\text{Large} \mid \text{Owned}) \uparrow$$

$\Rightarrow$  **fear**: the DE **amplifies**. Larger losses are rarer events  $\Rightarrow$  bigger marginal updates  $\Rightarrow$  stronger reinforcement.

## What the model buys us

A *single* representativeness-plus-skewness channel reproduces (i) the baseline DE, (ii) attenuation by wins, and (iii) amplification by losses – without invoking realization utility, sliding reference points, or distinct utility parameters per investor.



# Who is most responsive?

The memory channel predicts greater susceptibility for investors who *rely more on prior episodes*: less risk-tolerant, younger, less wealthy. We test all three.

| Cut                             | fear ( $\text{Gain} \times \text{LossExp}$ ) | greed ( $\text{Gain} \times \text{WinExp}$ ) |
|---------------------------------|----------------------------------------------|----------------------------------------------|
| Risk-averse vs. risk-seeking    | stronger for risk-averse                     | stronger for risk-averse                     |
| Younger ( $\leq 36$ ) vs. older | +0.215*** vs. +0.027*                        | -0.115*** vs. -0.028***                      |
| Lower wealth vs. higher         | stronger for lower-wealth                    | stronger for lower-wealth                    |

**Reading.** The asymmetry is concentrated where memory-based heuristics are most likely to bind: among less-experienced, less-wealthy, less risk-tolerant investors. Risk-seeking investors react to wins but not losses.

**Demographic gradient is exactly what a memory account predicts – and what a static-preference account does not.**

## Wide framing: portfolio-level evidence

Add Portfolio Gain (= 1 if account is net positive at the decision) and its interactions:

|                                                | Losing-experience block | Winning-experience block |
|------------------------------------------------|-------------------------|--------------------------|
| Gain                                           | 5.34***                 | 5.30***                  |
| Portfolio Gain $\times$ Gain                   | -8.35***                | -8.10***                 |
| Gain $\times$ Exp                              | -0.0591***              | -0.0663***               |
| Portfolio Gain $\times$ Exp                    | -0.1215***              | -0.0773***               |
| PfG $\times$ Gain $\times$ LossExp <b>fear</b> | +0.3057***              | -                        |
| PfG $\times$ Gain $\times$ WinExp <b>greed</b> | -                       | +0.1056***               |

**Panel B.** With Portfolio Gain = Gain = 1: net effect of one additional salient *loss*  $\approx +0.125$ ; one additional salient *win*  $\approx -0.038$ .

**Even after fully embedding decisions in their portfolio context, the asymmetry survives.**

# Taking stock

1. **Experience matters – asymmetrically.** Salient *wins* attenuate the disposition effect; salient *losses* amplify it.
2. Asymmetry is robust to thresholds, holding periods, portfolio context, mechanical bias, and demographic cuts. It *sharpens* with salience.
3. A **cued-recall memory model** – temporal context plus return skewness – produces the baseline DE *and* the asymmetric experience response from a single mechanism. Same channel: *overoptimism, extrapolative buying*, weaker DE for *mutual funds*.
4. Heterogeneity (younger, less-wealthy, more risk-averse investors most responsive) is what the memory account predicts.

## What it means

The disposition effect is not a fixed individual trait. It is a **state variable** shaped by the history of realized outcomes an investor carries into the next decision – through memory, not through shifting preferences.

For investors and advisors: salient losses don't make people more cautious – they sharpen the urge to lock in gains. *Fear breeds greed.*

# Thank you

Tai Lo Yeung · Rong Liu · Jessica Wachter · Michael Kahana · Yongjie Zhang

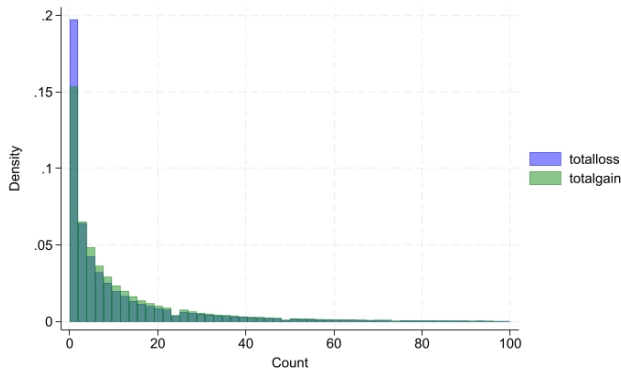
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SFS Cavalcade 2026 – comments and suggestions warmly welcome.

# Appendix

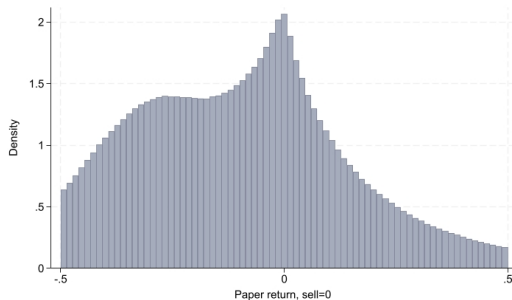
## Appendix | Distribution of trading experience



Per-account cumulative count of winning (green) and losing (purple) episodes at end of sample. Both are highly right-skewed.

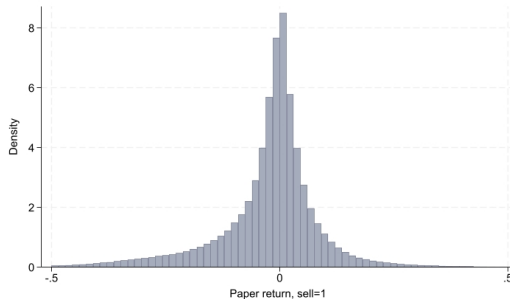
The mass at “one loss” reflects investors who exit after their first painful trade; beyond that, wins and losses have similar tail shapes.

### Paper (unrealized) returns



Heavy right tail; skewness = 23.1.

### Realized returns



Much closer to symmetric – investors realize their gains.

**The skewness gap is the disposition effect, visualised.**

## Appendix | The 30% threshold: asymmetry magnified

| Dep. var.: Sale $\times$ 100      | (1) Counts only | (2) +Interactions |
|-----------------------------------|-----------------|-------------------|
| Gain                              | 2.51***         | 2.43***           |
| Loss experience                   | 0.091*          | 0.066             |
| Win experience                    | -0.004          | 0.015             |
| Gain $\times$ LossExp <b>fear</b> |                 | <b>+0.091***</b>  |
| Gain $\times$ WinExp <b>greed</b> |                 | <b>-0.063*</b>    |
| Inv/Stock/Time FE                 | Yes             | Yes               |
| Observations                      | 43,388,866      | 43,388,866        |

At 30%, each significant loss raises selling propensity for winners by  $\approx 9\times$  more per event than at 5% – and *each* extra big loss adds nearly a full percentage point to the DE on its own.

**Salience scales the asymmetry, exactly as the memory account predicts.**



# Appendix | Mechanical bias: Monte Carlo

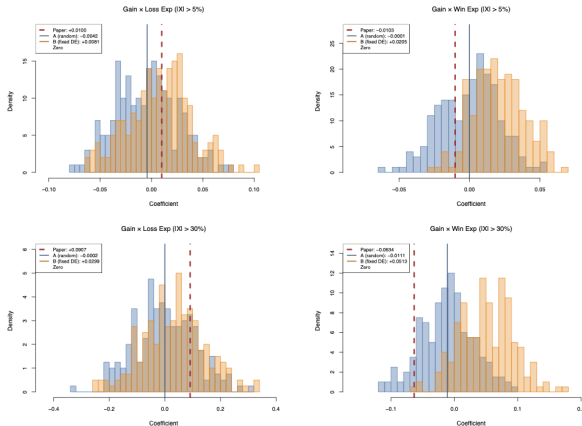
## Design.

5,000 simulated investors, 35 sells each, 200 reps. Three nulls:

- A. Random selling
- B. Fixed DE (winners  $1.5 \times$  losers, independent of experience)
- C. B + V-shape (Ben-David & Hirshleifer 2012)

## Reading.

Under every null, the average simulated interaction is *statistically zero or pushes against our finding*. Our  $(+0.010, -0.010)$  pair is not produced in a single replication.



Dashed red: paper estimate. Blue: random null. Orange: fixed-DE null.

Portfolio arithmetic cannot generate the asymmetry.

# Appendix | Memory model – math details

**Step 1. Memory matrix.** Episodes  $(f_i, x_i)$ :

$$M_n = M_0 + \sum_{i=1}^n x_i f_i^\top.$$

**Step 2. Cued recall.** Cue  $f_{\text{cue}}$  retrieves context, then features:

$$x_{\text{cue}} \propto M f_{\text{cue}}, \quad \tilde{f} \propto M^\top x_{\text{cue}}.$$

**Step 3. Competitive recall** (Kahana et al. 2025):

$$\mathbb{P}(\text{recall } f_i) \propto (x_i \cdot x_{\text{cue}}) \cdot \|f_i \cdot \tilde{f}\|_0.$$

With orthogonal-context evolution and shared-feature suppression of repeats, this reduces to a normalized co-occurrence:

$$\mathbb{P}(\text{Large} \mid \text{Owned}) \propto \frac{n_{\text{Owned}, \text{Large}}}{\sqrt{n_{\text{Owned}} n_{\text{Large}}}}.$$

**Step 4. Experience updating.** A winning experience replaces  $n_{\text{Owned}, \text{Large}} \mapsto n_{\text{Owned}, \text{Large}} - k$  (and adds to Sold), monotonically reducing  $\mathbb{P}(\text{Large} \mid \text{Owned})$ . A losing experience strips small losses from Owned, raising the relative weight on Large.

**One mechanism, one set of primitives – the asymmetric experience effect drops out.**

The model implies the DE strength scales with two structural features of the asset.

1. **Right-skewness of returns.** Large gains dominate retrieval under “stocks I own” precisely because individual-stock returns are heavily right-skewed (skewness  $\approx 23$  in our data).
2. **Directness of the decision.** Self-attributed trading and ownership decisions encode richer memory traces with the “Owned” cue.

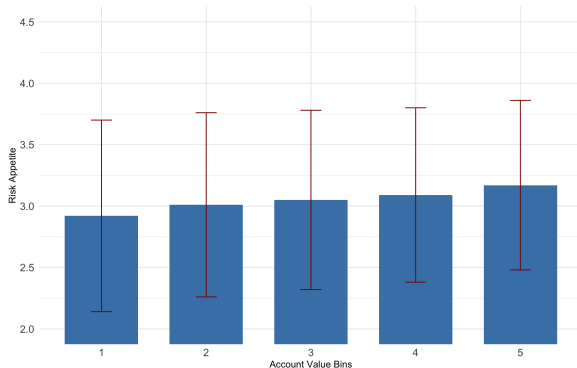
Mutual funds score lower on both.

- Diversification flattens the return distribution  $\Rightarrow$  kills the right tail that drives gain-dominant recall.
- Trading is *delegated* to the fund manager  $\Rightarrow$  weaker self-attribution of losses.

**Prediction: weaker, and at the limit reversed, DE for mutual funds.**

**Consistent with the data.** Chang, Solomon & Westerfield (JF 2016) document, within the same investors, a positive DE for individual stocks and a negative DE ( $-5$  pp) for mutual funds, with a delegation/blame-attribution interpretation that maps cleanly onto cued recall. Corroborating: Bailey, Kumar & Ng (2011); Ivković & Weisbenner (2009); Calvet, Campbell & Sodini (2009).

## Appendix | Wealth, risk, and the asymmetry

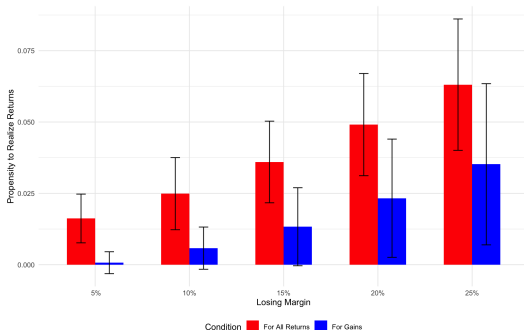


Average self-reported risk appetite (1=most averse, 5=most seeking) by account-value quintile,  $\pm 1$  SD. Risk and wealth co-move.

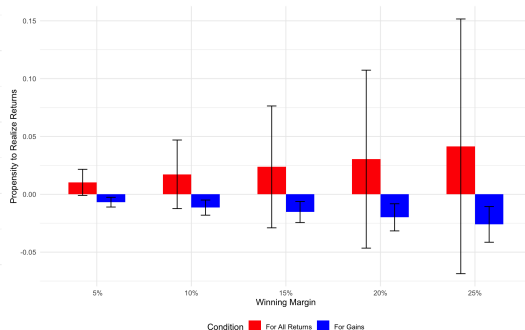
Wealth and risk-tolerance jointly buffer the asymmetric experience effect: the risk-seeking, wealthy tail responds to wins but not losses; the risk-averse, lower-wealth tail responds strongly to both, with the loss-amplification pattern intact.

## Appendix | Where the fear interaction comes from

### Losing margin



### Winning margin



Decomposition of  $P(\text{realize})$  across margin cutoffs, separately for all returns vs. gains only.

**Losses lift the propensity to realize *everything*, but especially gains.**